

Report 4: Training and inference panels: Mehregan, Nguyen, Ravivarapu

Prepared by Devin Auch
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Enhancing Adaptive Deep Brain Stimulation via Efficient Reinforcement Learning (Mehregan et al.)

Closed-Loop Neuromorphic Deep Brain Stimulation (Nguyen et al.)

Sample-Efficient Reinforcement Learning Controller for Deep Brain Stimulation in Parkinson's Disease (Ravivarapu et al.)

Overview (August 30, 2026): Mehregan et al. replication is fully complete with all figure panels passing quantitative gates. **Ravivarapu et al.** has all training, frequency inference, quantization, and ablation panels (Figs 4a–7) passing quantitative gates. **Nguyen et al.** Figs 3, 4, and 5 (biomarker distribution, 500-episode training dynamics, and CBGT spike count / DBS energy scaling) are all passing quantitative gates (Figs 6–7 to follow).

Same caveats as prior reports: replications from paper text and figures (no released training / environment source), shared plant = Python reconstruction of [Kumaravelu et al. \(2016\)](#). All replication figures include digitized ground-truth curves extracted directly from the source papers to enforce hard quantitative gates. Full gates and run commands: [figures/mehregan/replications.md](#), [figures/nguyen/replications.md](#), [figures/ravivarapu/replications.md](#).

Compute upgrade: AMD Ryzen 7 5800X (8c/16t) → Ryzen 9 5900XT (16c/32t)

Report 3 listed a workstation compute upgrade as upcoming work. That upgrade is **done** (August 2026): the training workstation is now **AMD Ryzen 9 5900XT (16 cores / 32 threads)**, replacing the previous **AMD Ryzen 7 5800X (8 cores / 16 threads)** that carried most replication trains since Report 1.

Why it matters for this project

One train at a time is not much faster. Each replication run caps in-process plant math at **1 thread** (OMP_NUM_THREADS=1, etc.), so a single Mehregan/Nguyen/Ravivarapu episode still spends ~the same wall-clock on the biophysical integrate loop. Because the neural ODE plant integration is pure floating-point math, one simulation run fully saturates one physical core's arithmetic execution pipeline. The upgrade buys **dedicated physical core concurrency and parallel sweep capacity**, not a magic speedup on one isolated 500-episode job.

Area	Before: Ryzen 7 5800X (8 cores / 16 threads)	Now: Ryzen 9 5900XT (16 cores / 32 threads)
Parallel worker capacity	Reserving 2 cores for OS, desktop, and tooling left 6 concurrent simulation workers max. A 4-variant ablation panel consumed nearly all available capacity.	Reserving 2–4 cores for OS, desktop, and tooling leaves 12–14 dedicated simulation workers running at 100% full speed without SMT thread contention.
Multi-seed / ablation sweeps	Running a 4-variant ablation across 3 seeds (12 runs total) required queuing across multiple serial batches.	Run an entire 12-run sweep (e.g., all 4 ablation variants across 3 seeds simultaneously) in a single parallel batch.
Multitasking & eval	Running continuous parameter sweeps alongside local evaluation and plotting scripts caused UI sluggishness and process contention.	Headroom to spare: keep 12 full simulation sweeps running in the background while compiling figures, analyzing data, and running desktop tools.
Calendar time on sweeps	Nguyen Fig. 4: each full train still hours; hyperparameter tuning waited on serial completion.	More parallel sweep lanes → more completed trains per week, not faster individual trains.

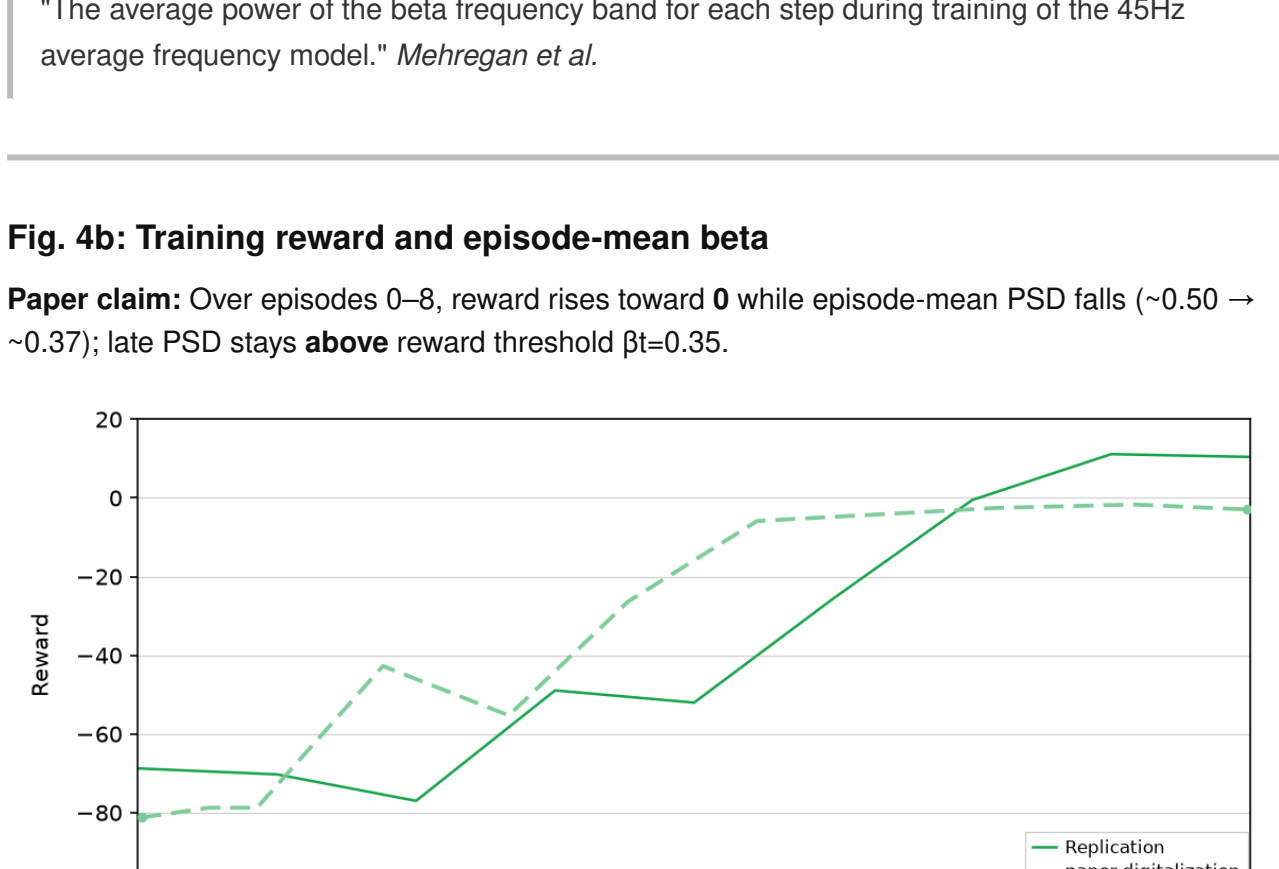
Delta since Report 3 (August 10)

Area	Report 3	Now
Compute	Planned CPU upgrade	Done: Ryzen 7 5800X (8c/16t) → Ryzen 9 5900XT (16c/32t)
Mehregan 4a–4b	Revisit in progress (digitization revisit; τ 3→1.4)	Pass: actor learning rate 7.5e-4, critic warmup steps 50, continuous carry
Nguyen 4	In progress (shape gates)	Pass: 500-episode reward and episode length trajectory gates
Nguyen 5	Not in Report 3 gallery	Pass: CBGT spike counts and DBS energy scaling (new in this report)
Ravivarapu 4a	In progress (shape gates)	Pass: full digitization trajectory gates
Ravivarapu 4b–7	Not in Report 3 gallery	Pass: training reward, 50/30 Hz inference, FP16 PTQ, ablation (new in this report)

Gallery: Mehregan et al.

Fig. 4a: Training beta vs step

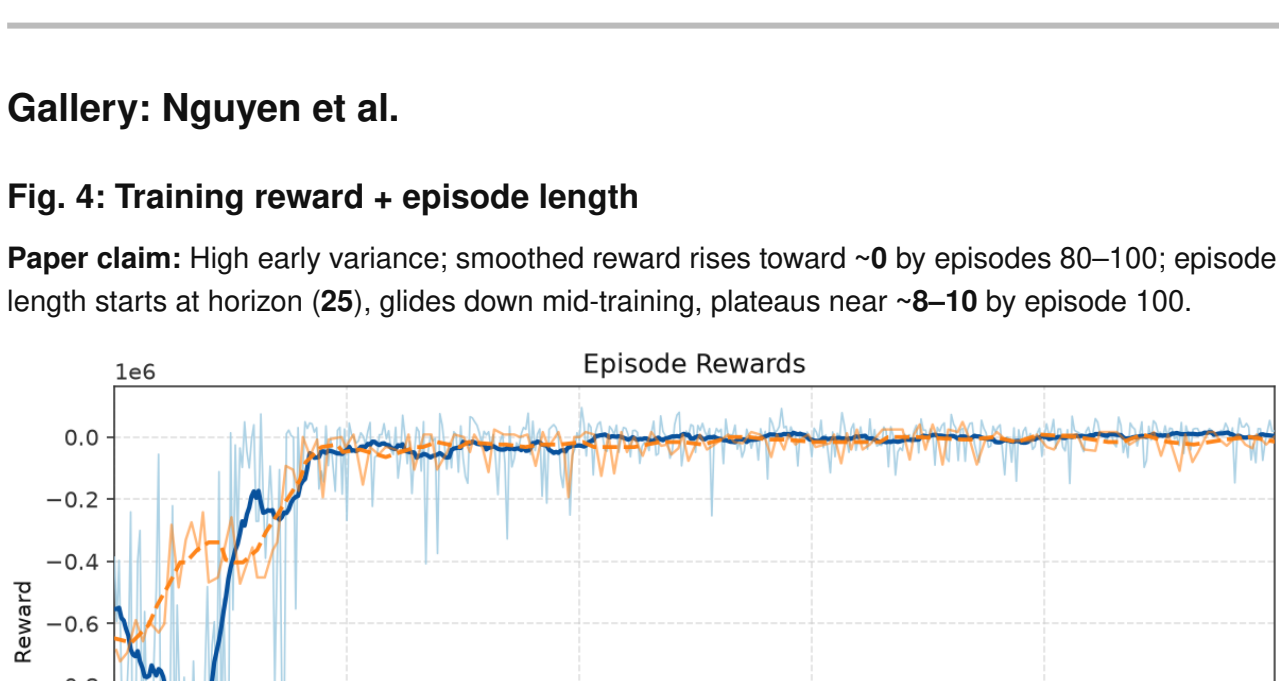
Paper claim: Noisy early β , sharp mid-run drop (~steps 130–150), lower late plateau during **45 Hz** DDPG training.



"The average power of the beta frequency band for each step during training of the 45Hz average frequency model." *Mehregan et al.*

Fig. 4b: Training reward and episode-mean beta

Paper claim: Over episodes 0–8, reward rises toward **0** while episode-mean PSD falls (~0.50 → ~0.37); late PSD stays **above** reward threshold $\beta t=0.35$.

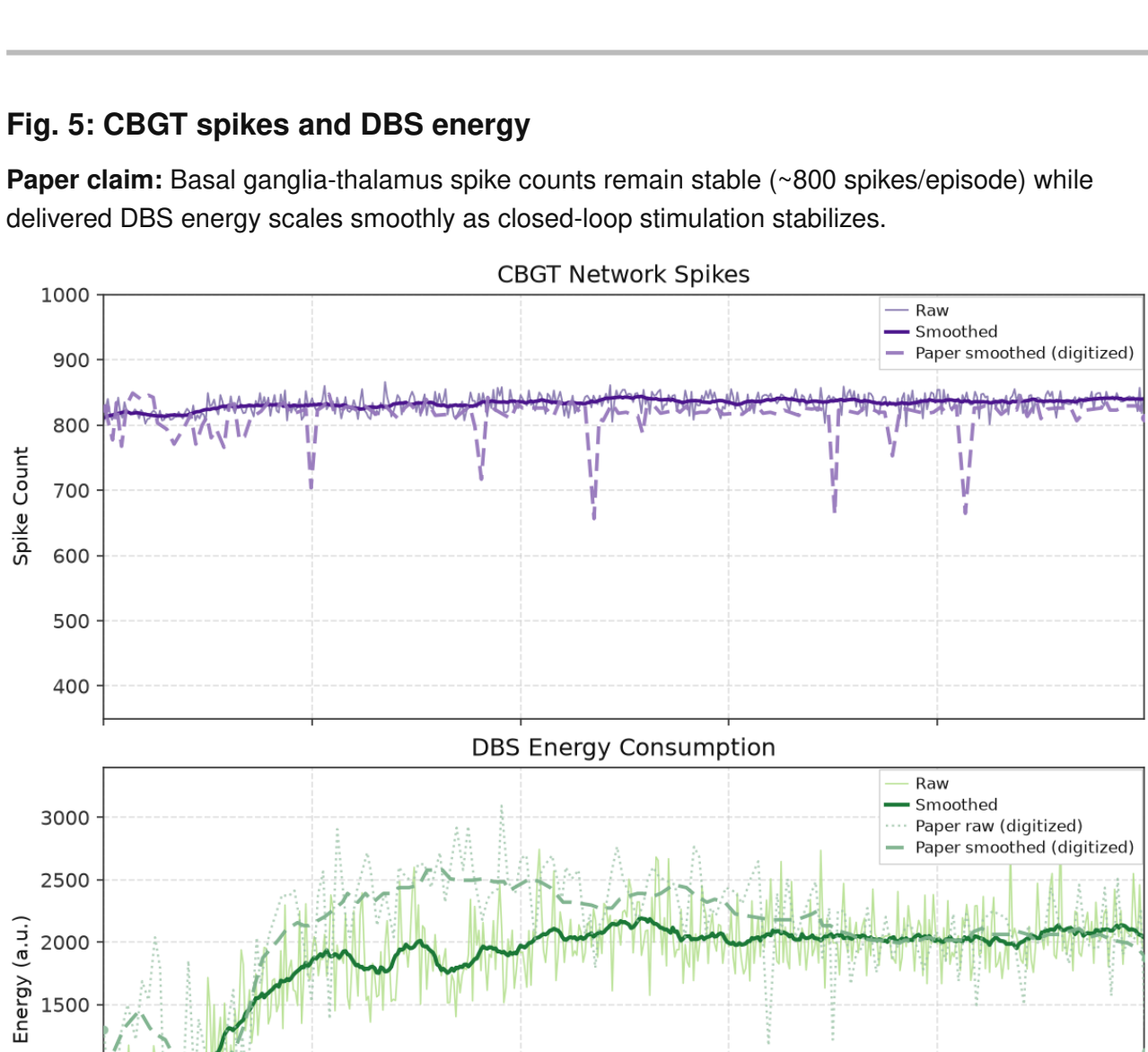


"Beta power in GPI during each episode of training of the 45Hz (top) and accumulated reward of each training episode for the 45Hz (bottom)." *Mehregan et al.*

Gallery: Nguyen et al.

Fig. 4: Training reward + episode length

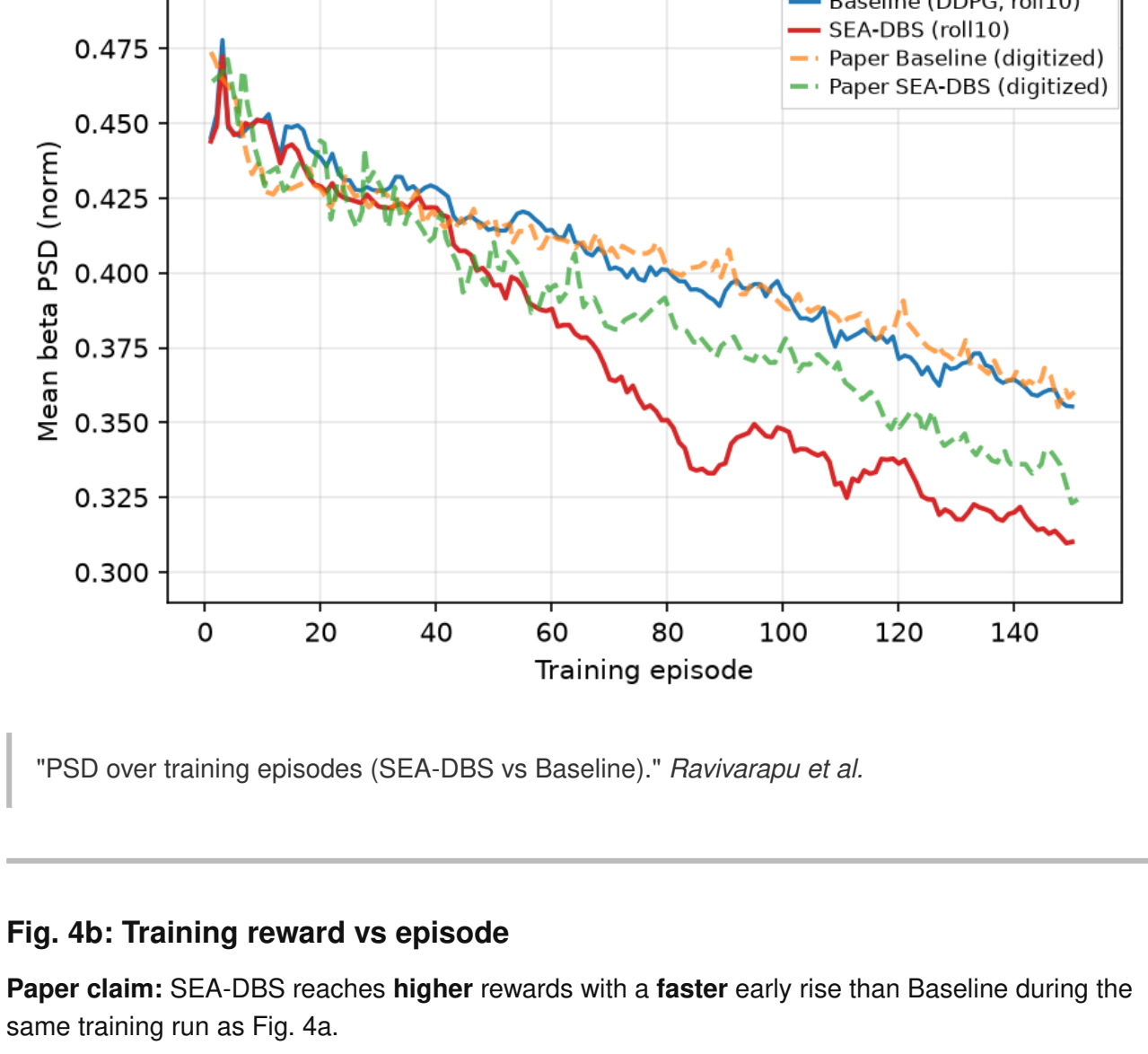
Paper claim: High early variance; smoothed reward rises toward **~0** by episodes 80–100; episode length starts at horizon (**25**), glides down mid-training, plateaus near **~8–10** by episode 100.



"Training rewards and episode lengths over 500 episodes, showing progression from exploration to optimization." *Nguyen et al.*

Fig. 5: CBGT spikes and DBS energy

Paper claim: Basal ganglia-thalamus spike counts remain stable (~800 spikes/episode) while delivered DBS energy scales smoothly as closed-loop stimulation stabilizes.

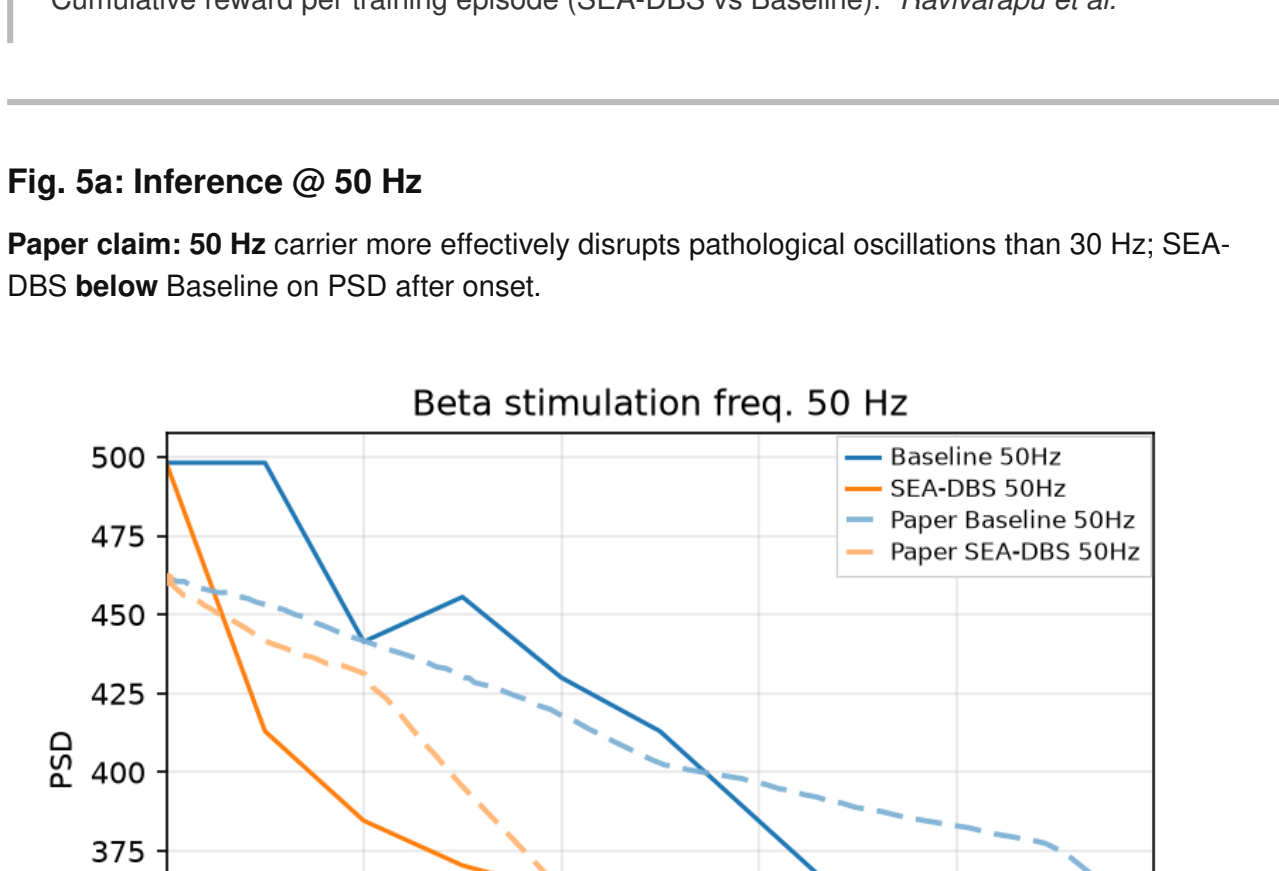


"Total number of spikes across the CBGT network (top) and total electrical energy delivered by DBS (bottom) per episode during training." *Nguyen et al.*

Gallery: Ravivarapu et al.

Fig. 4a: Training PSD (Baseline vs SEA-DBS)

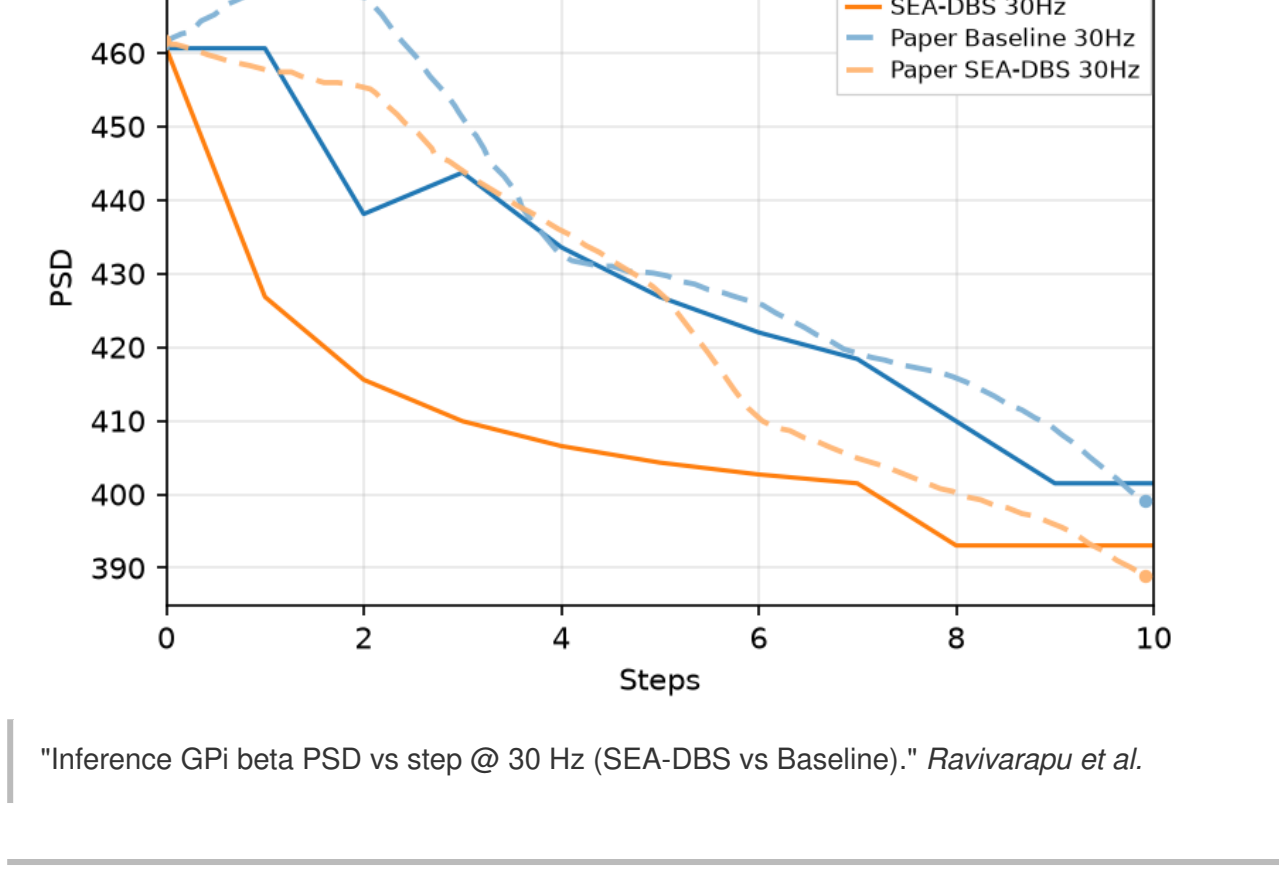
Paper claim: SEA-DBS shows stronger, more consistent β suppression over **150** episodes than Baseline DDPG.



"PSD over training episodes (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 4b: Training reward + episode

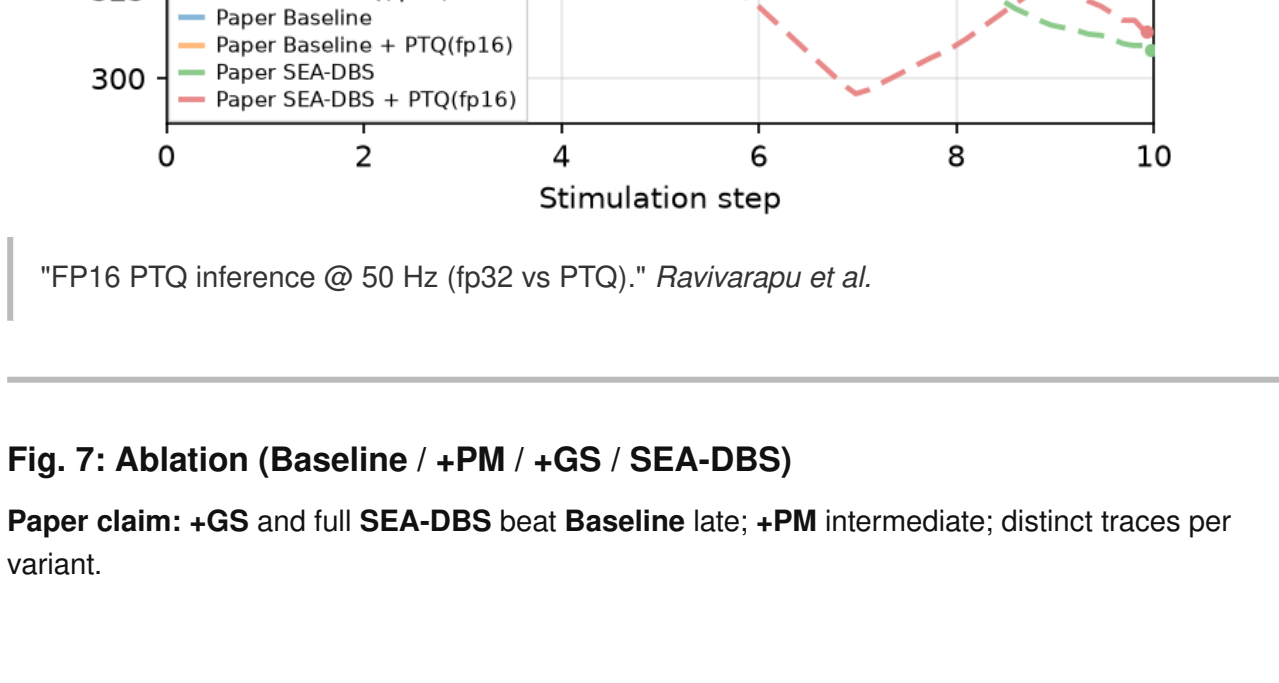
Paper claim: SEA-DBS reaches **higher** rewards with a **faster** early rise than Baseline during the same training run as Fig. 4a.



"Cumulative reward per training episode (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 5a: Inference @ 50 Hz

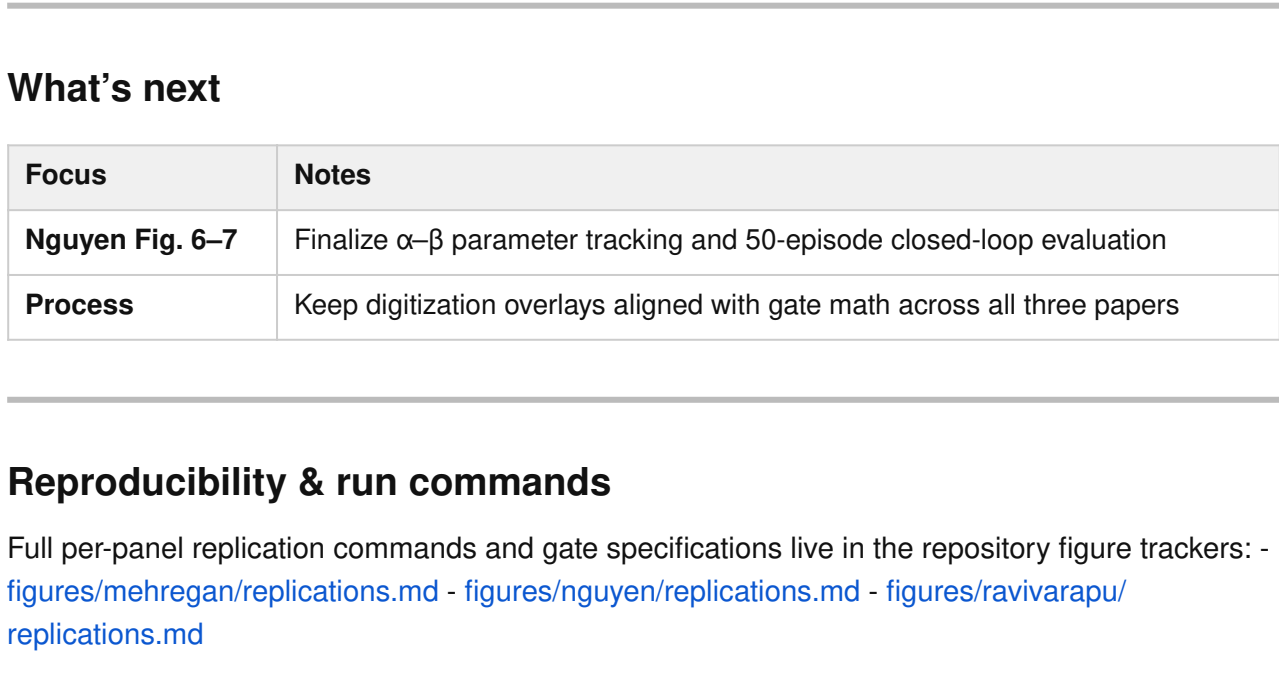
Paper claim: **50 Hz** carrier more effectively disrupts pathological oscillations than 30 Hz; SEA-DBS **below** Baseline on PSD after onset.



"Inference GPI beta PSD vs step @ 50 Hz (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 5b: Inference @ 30 Hz

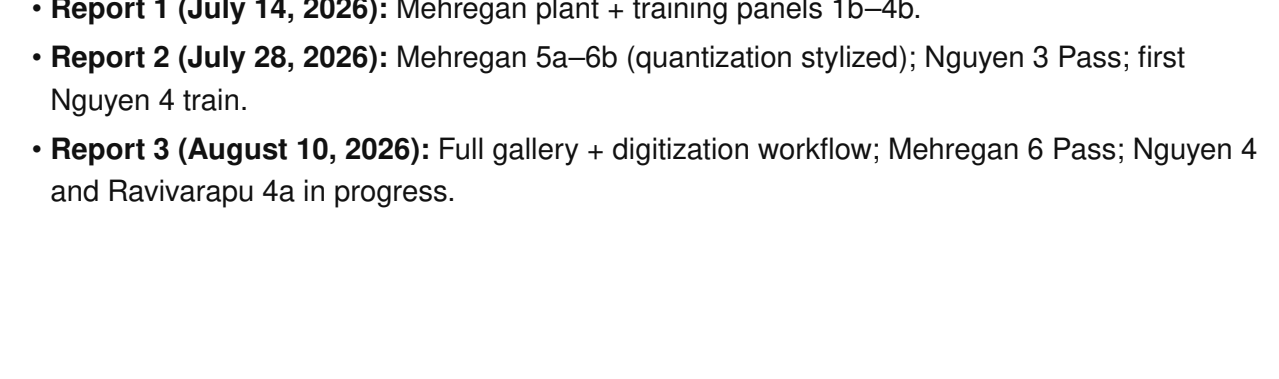
Paper claim: **30 Hz** less effective than 50 Hz; SEA-DBS still **beats** Baseline on PSD.



"Inference GPI beta PSD vs step @ 30 Hz (SEA-DBS vs Baseline)." *Ravivarapu et al.*

Fig. 6: FP16 PTQ @ 50 Hz

Paper claim: Post-train **fp32** vs **PTQ fp16** inference at 50 Hz; PTQ tracks fp32 suppression.



"FP16 PTQ inference @ 50 Hz (fp32 vs PTQ)." *Ravivarapu et al.*

Fig. 7: Ablation (Baseline / +PM / +GS / SEA-DBS)

Paper claim: **+GS** and full **SEA-DBS** beat **Baseline** late; **+PM** intermediate; distinct traces per variant.

"Ablation: Baseline, +PM, +GS, and full SEA-DBS @ 50 Hz." *Ravivarapu et al.*

What's next

Focus	Notes
Nguyen Fig. 6–7	Finalize α – β parameter tracking and 50-episode closed-loop evaluation
Process	Keep digitization overlays aligned with gate math across all three papers

Reproducibility & run commands

Full per-panel replication commands and gate specifications live in the repository figure trackers: - [figures/mehregan/replications.md](#) - [figures/nguyen/replications.md](#) - [figures/ravivarapu/replications.md](#)

```
git clone https://github.com/wilzen3476/r1-adaptive-dbs.git
cd r1-adaptive-dbs

# Mehregan Fig. 4a (DDPG 45 Hz training beta vs step)
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/mehregan/4a/plot.py

# Ravivarapu Fig. 7 (4-variant ablation sweep at 50 Hz)
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/ravivarapu/7/plot.py

# Replot any panel from cached simulation series without re-training
uv run python -m r1_adaptive_dbs.run scripts/figures/papers/ravivarapu/7/plot.py --plot-only
```

Environment note: All runs were executed and verified locally using uv. Standard installation and dependency management procedures are provided in `scripts/setup.sh` and the repository documentation.

Prior reports

- **Report 1 (July 14, 2026):** Mehregan plant + training panels 1b–4b.
- **Report 2 (July 28, 2026):** Mehregan 5a–6b (quantization stylized); Nguyen 3 Pass; first Nguyen 4 train.
- **Report 3 (August 10, 2026):** Full gallery + digitization workflow; Mehregan 6 Pass; Nguyen 4 and Ravivarapu 4a in progress.